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Hybrid wire localization length measurement device

Abstract

A superconductor-semiconductor device is provided, including a hybrid superconductor-semiconductor wire. The superconductor-semiconductor device may further include a hybrid localization length (LL) measurement device including a plurality of contact gates located above the hybrid superconductor-semiconductor wire in a thickness direction. The hybrid LL measurement device may further include a conductance sensor electrically coupled to the plurality of contact gates.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Patent Application Ser. No. 63/488,150, filed Mar. 2, 2023, the entirety of which is hereby incorporated herein by reference for all purposes.

BACKGROUND

(1) The Localization Length (LL) is the length scale that characterizes disorder strength in a low-dimensional (e.g., approximately one-dimensional) device when charge carriers such as electrons flow through the device. The LL therefore determines how conductance is reduced as the material sample size is increased and is a statistical measure of disorder that is used to characterize material systems and processing methods. In an approximately one-dimensional semiconductor wire, the LL defines how far charge carriers travel in a material before a scattering event.

SUMMARY

(2) According to one aspect of the present disclosure, a superconductor-semiconductor device is provided, including a hybrid superconductor-semiconductor wire. The superconductor-semiconductor device further includes a hybrid localization length (LL) measurement device including a plurality of contact gates located above the hybrid superconductor-semiconductor wire in a thickness direction. The hybrid LL measurement device further includes a conductance sensor electrically coupled to the plurality of contact gates.

(3) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 schematically shows components of a superconductor-semiconductor device at which LL measurement may be performed, according to one example embodiment.

(2) FIG. 2 schematically shows the superconductor-semiconductor device, including a plurality of gate electrodes, according to the example of FIG. 1.

(3) FIG. 3 schematically shows the superconductor-semiconductor device in an example in which the superconductor-semiconductor device includes a test measurement system and a topological quantum computing device, the test measurement system including a controller and a hybrid LL measurement device, according to the example of FIG. 1.

(4) FIG. 4A schematically shows an example top view of the superconductor-semiconductor device, including a plurality of plunger gates and a plurality of contact gates, according to the example of FIG. 2.

(5) FIG. 4B shows the example top view of the superconductor-semiconductor device when voltages are applied to the plunger gates and the contact gates, according to the example of FIG. 4A.

(6) FIGS. 4C-4E show example cross-sections of the superconductor-semiconductor device, according to the example of FIG. 4A.

(7) FIG. 5 schematically shows an example top-down view of another superconductor-semiconductor device at which the conductance values of a plurality of wire segments are configured to be measured in parallel, according to the example of FIG. 2.

(8) FIG. 6 shows a plot of the conductance of the hybrid superconductor-semiconductor wire as a function of a contact gate voltage, according to the example of FIG. 1.

(9) FIG. 7 shows a plot of the LL within a wire segment as a function of the voltage applied to a plunger gate that defines the wire segment, according to the example of FIG. 1.

(10) FIG. 8A shows a flowchart of a method for performing an LL measurement at a superconductor-semiconductor device, according to the example of FIG. 1.

(11) FIG. 8B shows additional steps of the method of FIG. 8A that may be performed when measuring conductance values.

(12) FIG. 9 shows a schematic view of an example computing system in which the controller of FIG. 3 may be instantiated.

DETAILED DESCRIPTION

(13) The LL is a quantity that is relevant to the functioning of topological quantum computing devices. Topological quantum computing devices use Majorana zero modes (MZMs) formed at boundaries between semiconducting and superconducting wires to form qubits. Under conditions in

which the LL is much shorter than the length of the semiconducting wire, an MZM does not form at the boundary between the semiconducting wire and the superconducting wire. MZM formation requires the LL within the semiconducting wire to be greater than the topological coherence length of the semiconductor. In addition, increasing the LL of the semiconductor leads to a reduction in the topological coherence length, and a shorter topological coherence length may allow for smaller device sizes. Thus, measurements of the LL may be used to assess the performance of a topological quantum computing device. For example, the LL of the semiconducting wire may be measured during the process of developing the quantum computing device. LL measurement may also be performed as a quality control test during quantum computing device manufacturing.

(14) For an electrically conducting wire in the localized regime, conductance scales with the wire length L and the LL ξ according to $\exp(-L/\xi)$. This scaling may be used to experimentally extract the LL by measuring the conductance in a series of wires with varying lengths, as discussed in further detail below. Exponential curve-fitting may then be used to estimate the LL.

(15) FIG. 1 schematically shows components of an example superconductor-semiconductor device **10** at which LL measurement may be performed. At the superconductor-semiconductor device **10** depicted in FIG. 1, a hybrid superconductor-semiconductor wire **20** is provided in a quantum well **18**. The quantum well **18** is located in a semiconductor layer **22** of the superconductor-semiconductor device **10**. The hybrid superconductor-semiconductor wire **20** may be formed by electrostatically confining the hybrid superconductor-semiconductor wire **20** wire in a two-dimensional electron gas (2DEG) that forms in the quantum well **18**. This electrostatic confinement produces a superconducting wire **20A** and a semiconducting wire **20B** arranged adjacent to and in parallel to each other to thereby form the hybrid superconductor-semiconductor wire **20**. The semiconductor layer **22** within which the quantum well **18** forms may, for example, be an indium arsenide (InAs) layer, and may be grown via molecular beam epitaxy.

(16) The superconductor-semiconductor device **10** depicted in FIG. 1 further includes a parent superconductor wire **14** located above and parallel to the hybrid superconductor-semiconductor wire **20** in the thickness direction **26**. The parent superconductor wire **14** may have a width of order 100 nm and a length of several microns. The parent superconductor wire **14** may, for example, be formed from aluminum or lead.

(17) The superconductor-semiconductor device **10** further includes a barrier layer **16** located between the parent superconductor wire **14** and the quantum well **18** in the thickness direction **26**. The barrier layer **16** may be included in the semiconductor layer. Accordingly, the barrier layer **16** acts as an energy barrier that separates the 2DEG formed in the quantum well **18** from the parent superconductor wire **14**. The barrier layer **16** may, for example, be an indium aluminum arsenide (InAlAs) layer.

(18) As depicted in the example of FIG. 1, the superconductor-semiconductor device **10** may further include a magnet **24** configured to produce an in-plane magnetic field B perpendicular to an axis of the hybrid superconductor-semiconductor wire **20**. Accordingly, the direction of the in-plane magnetic field B may be in the plane of the semiconductor layer **22** but perpendicular to a length direction of the hybrid superconductor-semiconductor wire **20** and the parent superconductor wire **14**. The strength of the in-plane magnetic field B may, for example, be approximately 1 T.

(19) FIG. 1 further shows a semiconducting buffer layer **23** located below the quantum well **18** in the thickness direction **26**. The semiconducting buffer layer **23** may be included in the semiconductor layer **22**. In addition, FIG. 1 shows a substrate layer **27** located below the semiconducting buffer layer **23**. The semiconducting buffer layer **23** and the substrate layer **27** may be located between the quantum well **18** and the magnet **24**. The magnet **24** may be spaced apart from the substrate layer **27**, as shown in FIG. 1. In other examples, the magnet **24** may be located at some other position, such as on a horizontal side of the superconductor-semiconductor device **10**.

(20) FIG. 1 further shows a gate electrode **12** located above the parent superconductor wire **14**. The gate electrode **12** may, for example, be an electrostatic gate electrode. Via the gate electrode **12**, a

voltage may be applied to the quantum well **18** to form the superconductor-semiconductor hybrid wire **20**. Applying the voltage to the gate electrode **12** may charge carriers in exposed semiconductor regions, which, in the example of FIG. **1**, are the regions of the quantum well **18** that are not underneath the parent superconductor wire **14**. The gate electrode **12** may be separated from the barrier layer **16** by an insulating layer **13**.

(21) As depicted in the example of FIG. **2**, the superconductor-semiconductor device **10** may include a plurality of gate electrodes **12** located above the parent superconductor wire **14** in the thickness direction **26**. FIG. **2** schematically shows the superconductor-semiconductor device **10** when the hybrid superconductor-semiconductor wire **20** forms at the quantum well **18**. The plurality of gate electrodes **12** shown in the example of FIG. **2** include a first contact gate **30A**, a second contact gate **30B**, and a plunger gate **32** located between the first contact gate **30A** and the second contact gate **30B**. The plurality of gate electrodes **12** may alternate between contact gates **30** and plunger gates **32** located along the axis of the hybrid superconductor-semiconductor wire **20**.

(22) In examples in which the semiconductor layer **22** is formed from an electron-like semiconductor, the plunger gate **32** may be set to a negative voltage relative to the semiconductor layer **22**. The negative voltage applied in such examples depletes charge carriers in regions of the semiconducting layer **22** that are not covered by the parent superconductor wire **14**, thereby forming the one-dimensional hybrid superconductor-semiconductor wire **20** in the semiconducting layer **22**.

(23) By applying voltages to the first contact gate **30A** and the second contact gate **30B**, respective contact regions **34** may be formed on the surface of the semiconductor layer **22**. These contact regions **34** may be formed by electrostatically charging the first contact gate **30A** and the second contact gate **30B**, thereby inducing opposite electrostatic charges on the surface of the semiconductor layer **22** within the contact regions **34**. The voltages applied to the first contact gate **30A** and the second contact gate **30B** may be opposite in sign to the charge on the plunger gate **32**. Thus, in the example discussed above in which a negative voltage is applied to the plunger gate **32**, positive voltages may be applied to the first contact gate **30A** and the second contact gate **30B**. Accordingly, a wire segment **36** of the hybrid superconductor-semiconductor wire **20** may be selected. The wire segment **36** is a portion of the hybrid superconductor-semiconductor wire **20** located between the first contact gate **30A** and the second contact gate **30B**. The wire segment **36** may have a length equal to the length of the plunger gate **32** located between the first contact gate **30A** and the second contact gate **30B**.

(24) FIG. **3** schematically shows the superconductor-semiconductor device **10** in an example in which the superconductor-semiconductor device **10** includes a test measurement system **40** and a topological quantum computing device **58**. In the example of FIG. **3**, the hybrid superconductor-semiconductor wire **20** is included in the topological quantum computing device **58**, which is coupled to the test measurement system **40** during performance testing. The test measurement system **40** shown in the example of FIG. **3** includes a hybrid LL measurement device **44** configured to measure the LL **54** of the hybrid superconductor-semiconductor wire **20**.

(25) The test measurement system **40** further includes a controller **42** configured to generate control instructions **56** that are executed at the hybrid LL measurement device **44**. For example, the control instructions **56** may include one or more bias voltage values **38** configured to be applied to the gate electrodes **12** included in the superconductor-semiconductor device **10**. In addition, the controller **42** is further configured to process measurement results obtained from the hybrid LL measurement device **44** to compute an LL **54**. The controller **42** may, for example, be a classical computing device including a processor and memory.

(26) The controller **42** is configured to control the hybrid LL measurement device **44** to measure a plurality of conductance values **50** of a plurality of wire segments **36** of the hybrid superconductor-semiconductor wire **20**. The plurality of wire segments **36** measured by the hybrid LL measurement device **44** have different respective lengths **52**. As discussed above, the lengths of the plurality of

wire segments **36** may be equal to the lengths of corresponding plunger gates **32** located above those wire segments **36** in the thickness direction **26**.

(27) The hybrid LL measurement device **44** shown in FIG. **3** includes a plurality of contact gates **30**, which may include the first contact gate **30A** and the second contact gate **30B** shown in FIG. **2**. As shown in the example of FIG. **3**, the plurality of contact gates **30** included in the hybrid LL measurement device **44** are each coupled to a conductance sensor **46**. The contact gates **30** may be coupled to the conductance sensor **46** via ohmic contacts located far from the hybrid superconductor-semiconductor wire **20** in order to avoid electromagnetic interference between the hybrid superconductor-semiconductor wire **20** and the conductance sensor **46**. The conductance sensor **46** is configured to output, to the controller **42**, respective conductance values **50** measured at the wire segments **36**.

(28) FIG. **4A** schematically shows an example top view of the superconductor-semiconductor device **10**. In the example of FIG. **4A**, a plurality of plunger gates **32** with different lengths are located above the parent superconductor wire **14**. The superconductor-semiconductor device **10** shown in FIG. **4A** further includes a plurality of contact gates **30** located above the parent superconductor wire **14** and interspersed with the plunger gates **32**. Thus, the contact gates **30** form wire segments **36** of three different lengths **52** in the example of FIG. **4A**. FIG. **4A** further shows a voltage source **68** that is coupled to the gate electrodes **12** by a voltage supply bus **70**. Via the voltage supply bus **70**, the voltage source **68** is configured to concurrently and independently apply respective voltages to the gate electrodes **12**.

(29) FIG. **4A** further shows a parent superconductor contact **60** located at an end of the parent superconductor wire **14**. The parent superconductor contact **60** is configured to act as a superconducting drain via which charge carriers may leave the parent superconductor wire **14** after tunneling into the parent superconductor wire **14**. Accordingly, the charge carriers that are removed from the parent superconductor wire **14** via the parent superconductor contact **60** do not contribute to the conductance measurement.

(30) FIG. **4B** shows the top view of the superconductor-semiconductor device **10** of FIG. **4A** when the voltage source **68** applies respective voltages to the gate electrodes **12** via the voltage supply bus **70**. In the example of FIG. **4B**, the controller **42** is configured to control the hybrid LL measurement device **44** to apply a first negative voltage **62** to the wire segment **36** via the corresponding plunger gate **32** when the conductance value **50** of the wire segment **36** is measured. The controller **42** may be configured to vary the first negative voltage **62** applied to the wire segment **36** when the conductance value **50** of the wire segment is measured. The first negative voltage **62** may be set to a plurality of different bias voltage values **38** included in the control instructions **56** during the conductance measurement.

(31) When the first negative voltage **62** is applied to the wire segment **36**, the parent superconductor wire **14** may at least partially screen the quantum well **18** from the first negative voltage **62** applied via the plunger gate **32**. The area of the semiconductor layer **22** that is screened from the first negative voltage **62** is the area of the semiconductor layer **22** at which the hybrid superconductor-semiconductor wire **20** forms. In contrast, portions of the semiconductor layer **22** on each side of the hybrid superconductor-semiconductor wire **20** in the plane of the semiconductor layer **22** may not be screened by the parent superconductor wire **14** from the first negative voltage **62** applied via the plunger gate **32**. Screening the area under the parent superconductor wire **14** without screening the other portions of the semiconductor layer **22** allows the charge carriers in the exposed regions of the semiconductor layer **22** to be depleted without depleting the carriers under the parent superconductor wire **14**.

(32) The controller **42** may be further configured to control the hybrid LL measurement device **44** to set the first contact gate **30A** and the second contact gate **30B** to a positive voltage **66** when the conductance value **50** of the wire segment **36** is measured. Thus, by applying the positive voltage **66** to the contact gates **30A** and **30B** on either side of a wire segment **36**, the hybrid LL

measurement device **44** is configured to use the contact gates **30A** and **30B** to select the wire segment **36** at which the conductance measurement is performed. Applying the positive voltage **66** to the first contact gate **30A** and the second contact gate **30B** allows charge carriers to flow along the hybrid superconductor-semiconductor wire **20** between the first contact gate **30A** and the second contact gate **30B**.

(33) In the example of FIG. **4B**, via the plunger gates **32** located above portions of the hybrid superconductor-semiconductor wire **20** other than the wire segment **36**, the controller **42** is configured to control the hybrid LL measurement device **44** to apply a second negative voltage **64** to those portions of the hybrid superconductor-semiconductor wire **20**. The magnitude of the second negative voltage **64** may be greater than that of the first negative voltage **62**. Accordingly, a tunnel barrier is formed between the plunger gates **32** and the portions of the hybrid superconductor-semiconductor wire **20** other than the wire segment **36**, thereby inhibiting the flow of charge carriers in such portions.

(34) By controlling which of the contact gates **30** are used to define the wire segments **36**, the controller **42** is configured to control the hybrid LL measurement device **44** to select wire segments **36** that have a plurality of different respective lengths. Thus, the controller **42** may be configured to iteratively select the contact gates **30** that are used as the first contact gate **30A** and the second contact gate **30B** to define the wire segments **36**.

(35) FIG. **4A** further shows cross-sections of the superconductor-semiconductor device **10** labeled as Section A, Section B, and Section C. Section A of the superconductor-semiconductor device **10** is shown in FIG. **4C**, according to one example. Section A is taken perpendicular to the parent superconductor wire **14**. In the example of FIG. **4C**, the semiconducting buffer layer **23**, the quantum well **18** (including the hybrid superconductor-semiconductor wire **20**), the barrier layer **16**, the insulating layer **13**, the parent superconductor wire **14**, and a plunger gate are shown in order from bottom to top in the thickness direction **26**. The magnetic field **B** points to the left in the example of FIG. **4C**.

(36) Section B of the superconductor-semiconductor device **10** is shown in FIG. **4D**. Section B is taken along the axis of the parent superconductor wire **14**. Similarly to Section A, Section B shows the semiconducting buffer layer **23**, the hybrid superconductor-semiconductor wire **20**, the barrier layer **16**, and the insulating layer **13** arranged from bottom to top in the thickness direction **26**. FIG. **4D** further shows a plunger gate **32** and two contact gates **30** located above the insulating layer **13** in the thickness direction **26**. In the example of FIG. **4D**, the magnetic field **B** points into the page.

(37) Section C of the superconductor-semiconductor device **10** is shown in FIG. **4E**. Section C passes through a contact region **34** and is taken perpendicular to the parent superconductor wire **14**. Section C shows the semiconducting buffer layer **23**, the quantum well **18**, the barrier layer **16**, the parent superconductor wire **14**, the insulating layer **13**, and a plunger gate **32** arranged from bottom to top in the thickness direction **26**. Section C further shows an ohmic contact **72** located between the barrier layer **16** and the insulating layer **13**. The ohmic contact **72** is located apart from the parent superconductor wire **14** in the plane of the barrier layer **16** in order to avoid electromagnetic interference between the ohmic contact **72** and the parent superconductor wire **14**.

(38) Returning to the example of FIG. **3**, the controller **42** is further configured to compute an LL **54** of the hybrid superconductor-semiconductor wire **20** based at least in part on the plurality of conductance values **50** and the plurality of lengths **52**. Since the conductance values **50**, the lengths **52**, and the LL **54** of the hybrid superconductor-semiconductor wire **20** approximately satisfy the relation $G \propto \exp(-L/\xi)$, where G is the conductance, the controller **42** may be configured to perform exponential curve-fitting to estimate the LL ξ . The scaling factor between G and $\exp(-L/\xi)$ is a device-dependent constant that is affected by the configuration of the superconductor-semiconductor device **10** and by the voltage applied to the hybrid superconductor-semiconductor wire **20**. This scaling factor is consistent among wire segments **36** of different lengths **52** included in the same hybrid superconductor-semiconductor wire **20**. The LL **54** of the hybrid

superconductor-semiconductor wire **20** may be measured as a function of a voltage applied to the gate electrode **12** that controls the potential in the semiconducting wire **20B** underneath the superconducting wire **20A**, as discussed in further detail below.

(39) Subsequently to estimating the LL **54** of the hybrid superconductor-semiconductor wire **20**, the controller **42** is further configured to output the LL **54**. The LL **54** may, for example, be output to a quality checking process that is executed to assess the performance of the topological quantum computing device **58**. The LL **54** may additionally or alternatively be output for display at a user interface.

(40) The hybrid LL measurement device **44** may enable independent benchmarking of the performance of superconductor-semiconductor devices **10** included in the topological quantum computing device **58**. The gate voltage of the gate electrode **12** may also be used to tune to the topological phase of the superconductor-semiconductor device when forming MZMs. Accordingly, the measurement of the LL **54** may utilize components already included in the topological quantum computing device **58** for other purposes, thereby decreasing the size and manufacturing complexity of the hybrid LL measurement device **44**.

(41) Accurately measuring the LL **54** of the hybrid superconductor-semiconductor wire **20** presents the following challenges: 1. Isolating a semiconductor contribution to the LL **54** from a superconductor contribution and other extraneous contributions. 2. Efficiently measuring multiple individual wires segments **36**, subject to the same processing conditions.

(42) At the hybrid LL measurement device **44**, the conductance contribution of the semiconducting wire **20B** may be isolated from that of the superconducting wire **20A**. The exponential relationship discussed above, according to which the conductance value **50** of a wire scales with the wire length **52** and the LL **54** as $G \propto \exp(-L/\xi)$, also holds for non-local conductance in a hybrid superconductor-semiconductor wire **20** that has a three-terminal (3T) configuration. The above relationship holds for bias voltage values **38** between an induced superconducting (SC) gap and a parent SC gap, since in this bias regime, quasiparticles in the semiconducting wire **20B** perform carrier transport. Thus, the LL **54** in the hybrid superconductor-semiconductor wire **20** may be measured by determining the length-dependence of non-local conductance in the above bias range. At bias voltage values **38** below the induced SC gap, no carrier transport occurs (ignoring supercurrent contributions). At bias voltage values **38** above the parent SC gap, carriers are also transported via the parent superconductor wire **14**. Thus, in a 3T device, approximately all the current flowing through the hybrid superconductor-semiconductor wire **20** is drained into the superconducting terminal above the parent SC gap.

(43) In order to approximately maximize the bias range over which non-local conductance may be measured, an in-plane magnetic field **B** may be applied perpendicular to the direction of the hybrid superconductor-semiconductor wire **20** using the magnet **24**, as discussed above. In other examples, the in-plane magnetic field **B** may be applied in a direction parallel to the hybrid superconductor-semiconductor wire **20**.

(44) In the hybrid superconductor-semiconductor wire **20**, there may be an additional current besides the quasi-particle current through the semiconducting wire **20B**. This additional current, referred to as the supercurrent, is a length-independent current through the superconducting wire **20A**. Since the presence of this supercurrent would interfere with the measurement of the LL **54**, the hybrid LL measurement device **44** is configured to suppress the supercurrent via two approaches. These methods of suppression are based on suppression of Andreev scattering at the superconductor-semiconductor interface within the hybrid superconductor-semiconductor wire **20**. First, the supercurrent is suppressed via the magnetic field **B**, which reduces Andreev reflection. Second, small junctions are used in the components of the superconductor-semiconductor device **10** that operate in high-conductance regimes. In the high-conductance regime, normal transmission through the junctions dominates over Andreev reflection at the junctions, thereby resulting in negligible supercurrent.

(45) Bias voltage values **38** within the range between 0 and the parent superconducting gap (typically several hundred μV) may be used when measuring the non-local conductance values **50**. Since the LL **54** depends on the energy of the quasiparticles, the measured LL **54** depends on the bias voltage value **38**. Performing the LL measurement at zero bias voltage results in an LL measurement with high accuracy. Alternatively, the conductance may be measured across the range of bias voltage values **38** between 0 and the parent SC gap voltage. The length-scaling of the average conductance over that range may be used to estimate LL. Compared to a measurement of the conductance at a bias voltage value of 0, a measurement of the conductance over the range between 0 and the parent SC gap has reduced noise but slightly overestimates the LL **54**.

(46) FIG. 5 schematically shows an example top-down view of another superconductor-semiconductor device **100** at which LL measurement is configured to be performed. In the example of FIG. 5, the controller **42** may be configured to control the hybrid LL measurement device **44** to measure respective conductance values **50** of a plurality of wire segments **36** in parallel. In the configuration of FIG. 5, the plunger gates **32** and the contact gates **30** each extend across a plurality of parent superconductor wires **14** arranged in parallel on the surface of the barrier layer **16**. As shown in the example of FIG. 5, the superconductor-semiconductor device **100** includes multiple different lengths of plunger gates **32**. Each of those plunger gates **32** forms multiple wire segments **36** under respective portions of the plurality of parent superconductor wires **14**. Within the set of wire segments **36** formed using each of the plunger gates **32**, charge carriers may be configured to flow between the wire segments **36** included in that set by traveling through the contact gates **30**. Accordingly, the hybrid LL measurement device **44** may be configured to measure conductance for a plurality of wire segments **36** collectively.

(47) The selection of the number of wire segments **36**, as well as the lengths **52** of those wire segments **36**, included in the superconductor-semiconductor device **10** is discussed below. Including wire segments **36** with a wide range of lengths **52** may allow the controller **42** to measure a wide range of LL values. In addition, including a large number of wire segments **36** in the superconductor-semiconductor device **10** may allow the LL measurement to be performed with high fidelity due to having a large sample size. In some examples, the number of wire segments **36** is constrained by a number of electronic control lines included in the controller **42**, since the controller **42** is configured to individually control the wire segments **36**.

(48) The lengths **52** of the wire segments **36** may be chosen according to the range of potentially measurable values of the LL **54**. The endpoints of the LL value range may be indicated as $\xi_{\text{sub.min}}$ and $\xi_{\text{sub.max}}$. The length **52** of the shortest wire segment **36** may be approximately equal to $\xi_{\text{sub.min}}$. The length **52** of the longest wire segment **36** may be chosen such that when the LL **54** is equal to $\xi_{\text{sub.max}}$, the expected signal proportional to $e_{\text{sup}} - L/\xi_{\text{sup.max}}$ is higher than the noise floor of the conductance sensor **46**. Thus, when the LL **54** is equal to $\xi_{\text{sub.max}}$, the conductance value **50** is measurable with high probability.

(49) Given the above bounds on the range within which the LL **54** is measured, the lengths **52** of the wire segments **36** may be chosen such that the lengths **52** increase by approximately exponential increments. Accordingly, each wire segment **36** may be longer than a preceding wire segment **36** by approximately a factor of n . The value of n may be chosen such that the total number of wire segments **36** is compatible with the available number of electronic control lines included in the controller **42**. For example, to measure LLs **54** between approximately $1\ \mu\text{m}$ and $10\ \mu\text{m}$, wire segments **36** with lengths of $500\ \text{nm}$, $1.3\ \mu\text{m}$, $2.2\ \mu\text{m}$, $3\ \mu\text{m}$, and $4\ \mu\text{m}$ may be used.

(50) FIG. 6 shows experimental results that demonstrate uniformity of the conductance values **50** between different contact gates **30** on the same hybrid superconductor-semiconductor wire **20**. FIG. 6 shows a conductance plot **200** of the conductance of the hybrid superconductor-semiconductor wire **20** (in units of $e_{\text{sup}}/2h$) as a function of the contact gate voltage (in units of V). The conductance is plotted for three different contact gates **30C**, **30D**, and **30E** located at different wire segments **36** of the hybrid superconductor-semiconductor wire **20** that have the same length **52**. In

the example of FIG. 6, the conductance values **50** obtained for the contact gates **30C**, **30D**, and **30E** are averaged over a plurality of bias voltage values **38** that result in charging energies greater than the induced SC gap. The experimental results of FIG. 6 were taken in a regime in which plunger gates **32** are set to highly negative voltages, thereby fully depleting the carriers in the wire segments **36**.

(51) As shown in the conductance plot **200** of FIG. 6, the measured conductance values **50** are approximately consistent between the contact gates **30C**, **30D**, and **30E** as a function of the contact gate voltage when measured at contact gate voltage values above approximately 0 V. In addition, the conductance plot **200** shows that conductance is suppressed at charging energies below the induced SC gap of the parent superconductor wire **14**.

(52) FIG. 7 shows an LL plot **210** of the LL **54** within a wire segment **36** as a function of the first negative voltage **62** applied to the plunger gate **32** that defines the wire segment **36**. In the LL plot **210** of FIG. 7, the LL **54** exhibits a U-shaped curve as a function of the first negative voltage **62**. The value that the controller **42** outputs as the LL **54** may be a value of the LL **54** located in the basin of the U-shaped curve. Accordingly, by controlling the hybrid LL measurement device **44** to measure the conductance values **50** over a range of different bias voltage values **38** applied to the plunger gate **32**, the controller **42** may be configured to obtain a larger sample of conductance values **50** with which the LL **54** may be estimated.

(53) FIG. 8A shows a flowchart of a method **300** for use with a superconductor-semiconductor device **10**. At step **302**, the method **300** may include controlling a hybrid LL measurement device **44** to measure a corresponding plurality of conductance values **50** of a plurality of wire segments **36** of a hybrid superconductor-semiconductor wire **20** that have different respective lengths **52**. The hybrid superconductor-semiconductor wire **20** may, for example, be included in a topological quantum computing device **58**.

(54) The hybrid LL measurement device **44**, which is also shown schematically in the example of FIG. 8A, includes a plurality of contact gates **30** located above the hybrid superconductor-semiconductor wire **20** in a thickness direction **26**. The hybrid LL measurement device **44** further includes a conductance sensor **46** electrically coupled to the plurality of contact gates **30**. Each of the plurality of wire segments **36** is a portion of the hybrid superconductor-semiconductor wire **20** located between a first contact gate **30A** and a second contact gate **30B** of the plurality of contact gates **30**. In addition to the hybrid superconductor-semiconductor wire **20** and the contact gates **30**, the superconductor-semiconductor device **10** at which the method **300** is performed further includes a plurality of plunger gates **32** located above the hybrid superconductor-semiconductor wire **20** in the thickness direction **26**. The lengths **52** of the plurality of wire segments **36** are equal to lengths of corresponding plunger gates **32** of the plurality of plunger gates **32**.

(55) At step **304**, the method **300** further includes computing an LL **54** of the hybrid superconductor-semiconductor wire **20** based at least in part on the plurality of conductance values **50** and the plurality of lengths **52**. The LL **54** may be computed via exponential curve fitting using the relationship $G \propto \exp(-L/\xi)$. The proportionality constant between the conductance and the exponential may be dependent on the device geometry and the voltages applied to the gate electrodes **12**. At step **306**, the method **300** further includes outputting the LL **54**.

(56) FIG. 8B shows additional steps of the method **300** that may be performed when measuring the conductance values **50** at step **302**. At step **308**, the method **300** may further include applying an in-plane magnetic field **B** to the superconductor-semiconductor device **10** perpendicular to an axis of the hybrid superconductor-semiconductor wire **20**. The in-plane magnetic field **B** may be applied using a magnet **24** located below the hybrid superconductor-semiconductor wire **20** in the thickness direction **26**. The in-plane magnetic field **B** may suppress supercurrent and increase the bias voltage range over which conductance measurement may be performed.

(57) At step **310**, the method **300** may further include controlling the hybrid LL measurement device **44** to apply a first negative voltage **62** to the wire segment **36** via the corresponding plunger

gate **32** when the conductance value **50** of the wire segment **36** is measured. The hybrid LL measurement device **44** may be controlled to set the first negative voltage **62** to a single bias voltage value **38** or to measure the conductance over a range of bias voltage values **38**.

(58) At step **312**, the method **300** may further include setting the first contact gate **30A** and the second contact gate **30B** to a positive voltage **66** when the conductance value **50** of the wire segment **36** is measured. Setting the first contact gate **30A** and the second contact gate **30B** to the positive voltage **66** may allow charge carriers to flow along the wire segment **36** between the first contact gate **30A** and the second contact gate **30B**.

(59) At step **314**, via the plunger gates **32** located above portions of the hybrid superconductor-semiconductor wire **20** other than the wire segment **36**, the method **300** may further include controlling the hybrid LL measurement device **44** to apply a second negative voltage **64** to those portions of the hybrid superconductor-semiconductor wire **20**. The second negative voltage **64** may be greater in magnitude than the first negative voltage **62**. Using the second negative voltage **64**, flow of charge carriers in the portions of the hybrid superconductor-semiconductor wire **20** other than the wire segment **36** may be inhibited.

(60) At step **316**, the method **300** may further include controlling the hybrid LL measurement device **44** to vary the first negative voltage **62** applied to the wire segment **36** when the conductance value **50** of the wire segment **36** is measured. The first negative voltage **62** may be varied within a range of bias voltage values **38** between 0 V and a parent superconducting gap of the parent superconductor wire **14**.

(61) Using the devices and methods discussed above, the LL of a hybrid superconductor-semiconductor wire may be approximated. Such an approximation may be performed when testing components of a topological quantum computing device, such as during a quality checking process performed during manufacturing of the topological quantum computing device. The LL of the hybrid superconductor-semiconductor wire affects its topological coherence length, which may determine whether MZMs are capable of forming at the hybrid superconductor-semiconductor wire. The topological coherence length may also affect eligible ranges of component sizes for some components of the topological quantum computing device. Thus, the LL significantly affects topological quantum computing device performance.

(62) In some embodiments, the methods and processes described herein may be tied to a computing system of one or more computing devices. In particular, such methods and processes may be implemented as a computer-application program or service, an application-programming interface (API), a library, and/or other computer-program product.

(63) FIG. **9** schematically shows a non-limiting embodiment of a computing system **400** that can enact one or more of the methods and processes described above. Computing system **400** is shown in simplified form. Computing system **400** may embody the controller **42** described above and illustrated in FIG. **3**. Components of computing system **400** may be included in one or more personal computers, server computers, tablet computers, home-entertainment computers, network computing devices, video game devices, mobile computing devices, mobile communication devices (e.g., smart phone), and/or other computing devices, and wearable computing devices such as smart wristwatches and head mounted augmented reality devices.

(64) Computing system **400** includes a logic processor **402** volatile memory **404**, and a non-volatile storage device **406**. Computing system **400** may optionally include a display subsystem **408**, input subsystem **410**, communication subsystem **412**, and/or other components not shown in FIG. **9**.

(65) Logic processor **402** includes one or more physical devices configured to execute instructions. For example, the logic processor may be configured to execute instructions that are part of one or more applications, programs, routines, libraries, objects, components, data structures, or other logical constructs. Such instructions may be implemented to perform a task, implement a data type, transform the state of one or more components, achieve a technical effect, or otherwise arrive at a desired result.

(66) The logic processor may include one or more physical processors configured to execute software instructions. Additionally or alternatively, the logic processor may include one or more hardware logic circuits or firmware devices configured to execute hardware-implemented logic or firmware instructions. Processors of the logic processor **402** may be single-core or multi-core, and the instructions executed thereon may be configured for sequential, parallel, and/or distributed processing. Individual components of the logic processor optionally may be distributed among two or more separate devices, which may be remotely located and/or configured for coordinated processing. Aspects of the logic processor may be virtualized and executed by remotely accessible, networked computing devices configured in a cloud-computing configuration. In such a case, these virtualized aspects are run on different physical logic processors of various different machines, it will be understood.

(67) Non-volatile storage device **406** includes one or more physical devices configured to hold instructions executable by the logic processors to implement the methods and processes described herein. When such methods and processes are implemented, the state of non-volatile storage device **406** may be transformed—e.g., to hold different data.

(68) Non-volatile storage device **406** may include physical devices that are removable and/or built-in. Non-volatile storage device **406** may include optical memory, semiconductor memory, and/or magnetic memory, or other mass storage device technology. Non-volatile storage device **406** may include nonvolatile, dynamic, static, read/write, read-only, sequential-access, location-addressable, file-addressable, and/or content-addressable devices. It will be appreciated that non-volatile storage device **406** is configured to hold instructions even when power is cut to the non-volatile storage device **406**.

(69) Volatile memory **404** may include physical devices that include random access memory. Volatile memory **404** is typically utilized by logic processor **402** to temporarily store information during processing of software instructions. It will be appreciated that volatile memory **404** typically does not continue to store instructions when power is cut to the volatile memory **404**.

(70) Aspects of logic processor **402**, volatile memory **404**, and non-volatile storage device **406** may be integrated together into one or more hardware-logic components. Such hardware-logic components may include field-programmable gate arrays (FPGAs), program- and application-specific integrated circuits (PASIC/ASICs), program- and application-specific standard products (PSSP/ASSPs), system-on-a-chip (SOC), and complex programmable logic devices (CPLDs), for example.

(71) The terms “module,” “program,” and “engine” may be used to describe an aspect of computing system **400** typically implemented in software by a processor to perform a particular function using portions of volatile memory, which function involves transformative processing that specially configures the processor to perform the function. Thus, a module, program, or engine may be instantiated via logic processor **402** executing instructions held by non-volatile storage device **406**, using portions of volatile memory **404**. It will be understood that different modules, programs, and/or engines may be instantiated from the same application, service, code block, object, library, routine, API, function, etc. Likewise, the same module, program, and/or engine may be instantiated by different applications, services, code blocks, objects, routines, APIs, functions, etc. The terms “module,” “program,” and “engine” may encompass individual or groups of executable files, data files, libraries, drivers, scripts, database records, etc.

(72) When included, display subsystem **408** may be used to present a visual representation of data held by non-volatile storage device **406**. The visual representation may take the form of a graphical user interface (GUI). As the herein described methods and processes change the data held by the non-volatile storage device, and thus transform the state of the non-volatile storage device, the state of display subsystem **408** may likewise be transformed to visually represent changes in the underlying data. Display subsystem **408** may include one or more display devices utilizing virtually any type of technology. Such display devices may be combined with logic processor **402**, volatile

memory **404**, and/or non-volatile storage device **406** in a shared enclosure, or such display devices may be peripheral display devices.

(73) When included, input subsystem **410** may comprise or interface with one or more user-input devices such as a keyboard, mouse, touch screen, or game controller. In some embodiments, the input subsystem may comprise or interface with selected natural user input (NUI) componentry. Such componentry may be integrated or peripheral, and the transduction and/or processing of input actions may be handled on- or off-board. Example NUI componentry may include a microphone for speech and/or voice recognition; an infrared, color, stereoscopic, and/or depth camera for machine vision and/or gesture recognition; a head tracker, eye tracker, accelerometer, and/or gyroscope for motion detection and/or intent recognition; as well as electric-field sensing componentry for assessing brain activity; and/or any other suitable sensor.

(74) When included, communication subsystem **412** may be configured to communicatively couple various computing devices described herein with each other, and with other devices.

Communication subsystem **412** may include wired and/or wireless communication devices compatible with one or more different communication protocols. As non-limiting examples, the communication subsystem may be configured for communication via a wireless telephone network, or a wired or wireless local- or wide-area network. In some embodiments, the communication subsystem may allow computing system **400** to send and/or receive messages to and/or from other devices via a network such as the Internet.

(75) The following paragraphs discuss several aspects of the present disclosure. According to one aspect of the present disclosure, a superconductor-semiconductor device is provided, including a hybrid superconductor-semiconductor wire. The superconductor-semiconductor device further includes a hybrid localization length (LL) measurement device including a plurality of contact gates located above the hybrid superconductor-semiconductor wire in a thickness direction. The hybrid LL measurement device further includes a conductance sensor electrically coupled to the plurality of contact gates. The above features may have the technical effect of allowing conductance measurements to be performed to thereby estimate the LL of the hybrid superconductor-semiconductor wire.

(76) According to this aspect, the superconductor-semiconductor device may further include a controller configured to control the hybrid LL measurement device to measure a corresponding plurality of conductance values of a plurality of wire segments of the hybrid superconductor-semiconductor wire that have different respective lengths. Each of the plurality of wire segments may be a portion of the hybrid superconductor-semiconductor wire located between a first contact gate and a second contact gate of the plurality of contact gates. The controller may be further configured to compute an LL of the hybrid superconductor-semiconductor wire based at least in part on the plurality of conductance values and the plurality of lengths. The controller may be further configured to output the LL. The above features may have the technical effect of performing conductance measurements of the wire segments and determining the LL of the hybrid superconductor-semiconductor wire based on the results of those measurements.

(77) According to this aspect, the superconductor-semiconductor device may further include a plurality of plunger gates located above the hybrid superconductor-semiconductor wire in the thickness direction, wherein the lengths of the plurality of wire segments are equal to lengths of corresponding plunger gates of the plurality of plunger gates. The above features may have the technical effect of providing a structure with which voltages may be applied to components of the superconductor-semiconductor device.

(78) According to this aspect, the controller is further configured to control the hybrid LL measurement device to apply a first negative voltage to the wire segment via the corresponding plunger gate when the conductance value of the wire segment is measured. The above features may have the technical effect of forming the hybrid superconductor-semiconductor wire under a parent superconductor wire.

(79) According to this aspect, the first contact gate and the second contact gate may be set to a positive voltage when the conductance value of the wire segment is measured. The above features may have the technical effect of allowing charge carriers to flow along the hybrid superconductor-semiconductor wire.

(80) According to this aspect, via the plunger gates located above portions of the hybrid superconductor-semiconductor wire other than the wire segment, the controller may be configured to control the hybrid LL measurement device to apply a second negative voltage to those portions of the hybrid superconductor-semiconductor wire. The above features may have the technical effect of inhibiting the flow of charge carriers in the portions of the hybrid superconductor-semiconductor wire other than the wire segment.

(81) According to this aspect, the controller may be further configured to control the hybrid LL measurement device to vary the first negative voltage applied to the wire segment when the conductance value of the wire segment is measured. The above features may have the technical effect of obtaining multiple measurements of the conductance of the wire segment, thereby allowing more precise estimation of the LL.

(82) According to this aspect, the hybrid superconductor-semiconductor wire may be located in a quantum well. The quantum well may be located in a semiconductor layer of the superconductor-semiconductor device. The above features may have the technical effect of forming the hybrid superconductor-semiconductor wire in a 2DEG.

(83) According to this aspect, the superconductor-semiconductor device may further include a parent superconductor wire located above and parallel to the hybrid superconductor-semiconductor wire in the thickness direction. The superconductor-semiconductor device may further include a barrier layer located between the parent superconductor wire and the quantum well in the thickness direction. The above features may have the technical effect of forming the hybrid superconductor-semiconductor wire via screening of a portion of the quantum well by the parent superconductor wire.

(84) According to this aspect, the parent superconductor wire may at least partially screen the quantum well from the first negative voltage applied via the plunger gate. The above features may have the technical effect of forming the hybrid superconductor-semiconductor wire via screening of a portion of the quantum well by the parent superconductor wire.

(85) According to this aspect, portions of the semiconductor layer on each side of the hybrid superconductor-semiconductor wire in a plane of the semiconductor layer may not be screened by the parent superconductor wire from the first negative voltage applied via the plunger gate. The above features may have the technical effect of forming an approximately one-dimensional hybrid superconductor-semiconductor wire.

(86) According to this aspect, the superconductor-semiconductor device may further include a magnet configured to produce an in-plane magnetic field perpendicular to an axis of the hybrid superconductor-semiconductor wire. The above features may have the technical effect of suppressing supercurrent at the superconductor-semiconductor interface within the hybrid superconductor-semiconductor wire.

(87) According to this aspect, the hybrid superconductor-semiconductor wire may be included in a topological quantum computing device. The above features may have the technical effect of providing a structure at which Majorana zero modes may form within the topological quantum computing device.

(88) According to another aspect of the present disclosure, a method for use with a superconductor-semiconductor device is provided. The method includes controlling a hybrid LL measurement device to measure a corresponding plurality of conductance values of a plurality of wire segments of a hybrid superconductor-semiconductor wire that have different respective lengths. The hybrid LL measurement device includes a plurality of contact gates located above the hybrid superconductor-semiconductor wire in a thickness direction. The hybrid LL measurement device

further includes a conductance sensor electrically coupled to the plurality of contact gates. Each of the plurality of wire segments is a portion of the hybrid superconductor-semiconductor wire located between a first contact gate and a second contact gate of the plurality of contact gates. The method further includes computing an LL of the hybrid superconductor-semiconductor wire based at least in part on the plurality of conductance values and the plurality of lengths. The method further includes outputting the LL. The above features may have the technical effect of performing conductance measurements of the wire segments and determining the LL of the hybrid superconductor-semiconductor wire based on the results of those measurements.

(89) According to this aspect, the superconductor-semiconductor device may further include a plurality of plunger gates located above the hybrid superconductor-semiconductor wire in the thickness direction. The lengths of the plurality of wire segments may be equal to lengths of corresponding plunger gates of the plurality of plunger gates. The above features may have the technical effect of providing a structure with which voltages may be applied to components of the superconductor-semiconductor device.

(90) According to this aspect, the method may further include controlling the hybrid LL measurement device to apply a first negative voltage to the wire segment via the corresponding plunger gate when the conductance value of the wire segment is measured. The above features may have the technical effect of forming the hybrid superconductor-semiconductor wire under a parent superconductor wire.

(91) According to this aspect, the method may further include setting the first contact gate and the second contact gate to a positive voltage when the conductance value of the wire segment is measured. The method may further include, via the plunger gates located above portions of the hybrid superconductor-semiconductor wire other than the wire segment, controlling the hybrid LL measurement device to apply a second negative voltage to those portions of the hybrid superconductor-semiconductor wire. The above features may have the technical effect of allowing charge carriers to flow along the hybrid superconductor-semiconductor wire while inhibiting the flow of charge carriers in the portions of the hybrid superconductor-semiconductor wire other than the wire segment.

(92) According to this aspect, the method may further include controlling the hybrid LL measurement device to vary the first negative voltage applied to the wire segment when the conductance value of the wire segment is measured. The above features may have the technical effect of obtaining multiple measurements of the conductance of the wire segment, thereby allowing more precise estimation of the LL.

(93) According to this aspect, the method may further include applying an in-plane magnetic field to the superconductor-semiconductor device perpendicular to an axis of the hybrid superconductor-semiconductor wire. The above features may have the technical effect of suppressing supercurrent at the superconductor-semiconductor interface within the hybrid superconductor-semiconductor wire.

(94) According to another aspect of the present disclosure, a superconductor-semiconductor device is provided, including a hybrid superconductor-semiconductor wire located in a quantum well included in a semiconductor layer of the superconductor-semiconductor device. The superconductor-semiconductor device further includes a barrier layer located above the semiconductor layer in the thickness direction. The superconductor-semiconductor device further includes a parent superconductor wire located above the barrier layer in the thickness direction and parallel to the hybrid superconductor-semiconductor wire. The superconductor-semiconductor device further includes a hybrid localization length (LL) measurement device including a plurality of gate electrodes located above the parent superconductor wire in a thickness direction. The plurality of gate electrodes includes a plurality of contact gates and a plurality of plunger gates. The hybrid LL measurement device further includes a conductance sensor electrically coupled to the plurality of contact gates. The hybrid LL measurement device is further configured to measure a

corresponding plurality of conductance values of a plurality of wire segments of the hybrid superconductor-semiconductor wire that have different respective lengths. The lengths of the plurality of wire segments are equal to lengths of corresponding plunger gates of the plurality of plunger gates. The above features may have the technical effect of performing conductance measurements of the wire segments and determining the LL of the hybrid superconductor-semiconductor wire based on the results of those measurements.

(95) It will be understood that the configurations and/or approaches described herein are exemplary in nature, and that these specific embodiments or examples are not to be considered in a limiting sense, because numerous variations are possible. The specific routines or methods described herein may represent one or more of any number of processing strategies. As such, various acts illustrated and/or described may be performed in the sequence illustrated and/or described, in other sequences, in parallel, or omitted. Likewise, the order of the above-described processes may be changed.

(96) The subject matter of the present disclosure includes all novel and non-obvious combinations and sub-combinations of the various processes, systems and configurations, and other features, functions, acts, and/or properties disclosed herein, as well as any and all equivalents thereof.

Claims

1. A superconductor-semiconductor device comprising: a hybrid superconductor-semiconductor wire; and a hybrid localization length (LL) measurement device including: a plurality of contact gates located above the hybrid superconductor-semiconductor wire in a thickness direction; and a conductance sensor electrically coupled to the plurality of contact gates.
2. The superconductor-semiconductor device of claim 1, further comprising a controller configured to: control the hybrid LL measurement device to measure a corresponding plurality of conductance values of a plurality of wire segments of the hybrid superconductor-semiconductor wire that have different respective lengths, wherein each of the plurality of wire segments is a portion of the hybrid superconductor-semiconductor wire located between a first contact gate and a second contact gate of the plurality of contact gates; compute an LL of the hybrid superconductor-semiconductor wire based at least in part on the plurality of conductance values and the plurality of lengths; and output the LL.
3. The superconductor-semiconductor device of claim 2, further comprising a plurality of plunger gates located above the hybrid superconductor-semiconductor wire in the thickness direction, wherein the lengths of the plurality of wire segments are equal to lengths of corresponding plunger gates of the plurality of plunger gates.
4. The superconductor-semiconductor device of claim 3, wherein the controller is further configured to control the hybrid LL measurement device to apply a first negative voltage to the wire segment via the corresponding plunger gate when the conductance value of the wire segment is measured.
5. The superconductor-semiconductor device of claim 4, wherein the first contact gate and the second contact gate are set to a positive voltage when the conductance value of the wire segment is measured.
6. The superconductor-semiconductor device of claim 5, wherein, via the plunger gates located above portions of the hybrid superconductor-semiconductor wire other than the wire segment, the controller is configured to control the hybrid LL measurement device to apply a second negative voltage to those portions of the hybrid superconductor-semiconductor wire.
7. The superconductor-semiconductor device of claim 4, wherein the controller is further configured to control the hybrid LL measurement device to vary the first negative voltage applied to the wire segment when the conductance value of the wire segment is measured.
8. The superconductor-semiconductor device of claim 4, wherein: the hybrid superconductor-semiconductor wire is located in a quantum well; and the quantum well is located in a

semiconductor layer of the superconductor-semiconductor device.

9. The superconductor-semiconductor device of claim 8, further comprising: a parent superconductor wire located above and parallel to the hybrid superconductor-semiconductor wire in the thickness direction; and a barrier layer located between the parent superconductor wire and the quantum well in the thickness direction.

10. The superconductor-semiconductor device of claim 9, wherein the parent superconductor wire at least partially screens the quantum well from the first negative voltage applied via the plunger gate.

11. The superconductor-semiconductor device of claim 10, wherein portions of the semiconductor layer on each side of the hybrid superconductor-semiconductor wire in a plane of the semiconductor layer are not screened by the parent superconductor wire from the first negative voltage applied via the plunger gate.

12. The superconductor-semiconductor device of claim 1, further comprising a magnet configured to produce an in-plane magnetic field perpendicular to an axis of the hybrid superconductor-semiconductor wire.

13. The superconductor-semiconductor device of claim 1, wherein the hybrid superconductor-semiconductor wire is included in a topological quantum computing device.

14. A method for use with a superconductor-semiconductor device, the method comprising: controlling a hybrid localization length (LL) measurement device to measure a corresponding plurality of conductance values of a plurality of wire segments of a hybrid superconductor-semiconductor wire that have different respective lengths, wherein: the hybrid LL measurement device includes: a plurality of contact gates located above the hybrid superconductor-semiconductor wire in a thickness direction; and a conductance sensor electrically coupled to the plurality of contact gates; and each of the plurality of wire segments is a portion of the hybrid superconductor-semiconductor wire located between a first contact gate and a second contact gate of the plurality of contact gates; computing an LL of the hybrid superconductor-semiconductor wire based at least in part on the plurality of conductance values and the plurality of lengths; and outputting the LL.

15. The method of claim 14, wherein: the superconductor-semiconductor device further includes a plurality of plunger gates located above the hybrid superconductor-semiconductor wire in the thickness direction; and the lengths of the plurality of wire segments are equal to lengths of corresponding plunger gates of the plurality of plunger gates.

16. The method of claim 15, further comprising controlling the hybrid LL measurement device to apply a first negative voltage to the wire segment via the corresponding plunger gate when the conductance value of the wire segment is measured.

17. The method of claim 16, further comprising: setting the first contact gate and the second contact gate to a positive voltage when the conductance value of the wire segment is measured; and via the plunger gates located above portions of the hybrid superconductor-semiconductor wire other than the wire segment, controlling the hybrid LL measurement device to apply a second negative voltage to those portions of the hybrid superconductor-semiconductor wire.

18. The method of claim 16, further comprising controlling the hybrid LL measurement device to vary the first negative voltage applied to the wire segment when the conductance value of the wire segment is measured.

19. The method of claim 14, further comprising applying an in-plane magnetic field to the superconductor-semiconductor device perpendicular to an axis of the hybrid superconductor-semiconductor wire.

20. A superconductor-semiconductor device comprising: a hybrid superconductor-semiconductor wire located in a quantum well included in a semiconductor layer of the superconductor-semiconductor device; a barrier layer located above the semiconductor layer in a thickness direction; a parent superconductor wire located above the barrier layer in the thickness direction

and parallel to the hybrid superconductor-semiconductor wire; a hybrid localization length (LL) measurement device including: a plurality of gate electrodes located above the parent superconductor wire in a thickness direction, wherein the plurality of gate electrodes includes a plurality of contact gates and a plurality of plunger gates; and a conductance sensor electrically coupled to the plurality of contact gates, wherein: the hybrid LL measurement device is configured to measure a corresponding plurality of conductance values of a plurality of wire segments of the hybrid superconductor-semiconductor wire that have different respective lengths; and the lengths of the plurality of wire segments are equal to lengths of corresponding plunger gates of the plurality of plunger gates.
